

Research Statement

Shashank Singh

Quiver methods for module categories over non-semisimple tensor categories

Research theme. My work develops explicit quiver-theoretic models for module categories and module functors over finite tensor categories. The guiding principle is that tensor-categorical structure becomes more usable once it can be written in concrete linear-algebraic terms.

1. Introduction, Background, and Motivation

My research lies at the intersection of representation theory, Hopf algebras, and tensor categories. A central theme is to make abstract categorical representation theory more explicit and computable by developing quiver-theoretic models for module categories, bimodule categories, and functors between them. In classical representation theory of finite-dimensional algebras, quivers and path algebras provide a bridge between structural algebra and concrete linear algebra: they encode modules by diagrams of vector spaces and maps, and they often turn classification problems into combinatorial ones. One of the enduring lessons of this perspective, going back to Gabriel and the representation theory of finite-dimensional algebras, is that the right combinatorial model does not merely repack known structure; it makes it calculable.

Many of the same structural questions now arise in the broader setting of finite tensor categories. These categories, introduced systematically by Etingof and Ostrik [EO04], provide a natural framework for the non-semisimple representation theory of finite-dimensional Hopf algebras and quantum groups. In this setting, module categories play the role of categorical representations, while module functors and bimodule categories encode higher Morita-theoretic structure. Andruskiewitsch and Mombelli [AM07] gave a classification of module categories over finite-dimensional Hopf algebras in terms of comodule algebras. Montgomery and Schneider [MS01] initiated a detailed study of Hopf actions on finite-dimensional algebras, particularly involving Taft algebras and their doubles. Kinser and Walton [KW16], and later Kinser and Oswald [KO21], gave quiver-theoretic classifications of actions of pointed Hopf algebras and certain small quantum groups on path algebras. In a parallel direction, Etingof, Kinser, and Walton [EKW21] developed the theory of tensor algebras in finite tensor categories, providing a conceptual way to encode algebra objects and their module categories inside a tensor category.

Taken together, these developments suggest a natural next step: to build a genuinely combinatorial calculus for non-semisimple tensor categories analogous to the role quivers play for finite-dimensional algebras. In semisimple settings, work of Elias and Heng [EH24], and Heng and Stroiński [HS26], on fusion quivers shows how effective a good combinatorial model can be. In the non-semisimple world, however, there remains a noticeable gap between abstract classification and explicit realization. Even when module categories and functors are known to exist and satisfy powerful structural theorems, they are often described indirectly through algebra objects, comodule algebras, or categorical universal properties rather than through concrete linear-algebraic data.

The central question underlying my current work is therefore the following:

Guiding question. Can one describe module categories and module functors over a non-semisimple tensor category in a genuinely explicit quiver-theoretic way, so that tensor-categorical compatibility becomes a concrete linear-algebraic condition rather than an abstract coherence constraint?

My research shows that, for the Taft tensor category, the answer is yes. Taft algebras $T_\ell(q)$ are pointed Hopf algebras generated by a grouplike element and a skew-primitive element, with relations $g^\ell = 1$, $x^\ell = 0$, and $xg = qgx$ for a primitive ℓ -th root of unity q . They are among the simplest non-semisimple Hopf algebras. Their representation categories $\mathcal{T} := \text{Rep}(T_\ell)$ are finite tensor categories in the sense of [EO04, EGNO15].

For each divisor $d \mid \ell$, I construct an explicit standard exact left $\mathcal{T}$ -module category $\mathcal{M}_d$ in terms of a nilpotent cycle quiver. I then study right exact $\mathcal{T}$ -module functors between these module categories through kernels in Deligne products. The key point is that the extra requirement of $\mathcal{T}$ -linearity can be rewritten as an explicit balancing relation on a mixed quiver grid. That relation is strong enough to collapse the apparent two-dimensional kernel calculus to one-dimensional cyclic data.

This point of view places quivers at the center of a broader program. The immediate goal is to understand the module-category theory of the Taft tensor category concretely. The larger goal is to develop methods that can be transported to other finite tensor categories, especially non-semisimple ones for which abstract classification is known in principle but not yet explicit in practice. In that sense, my work sits at the intersection of quiver representations, Hopf algebras, and categorical Morita theory.

2. Current Work and Results

The philosophy from the introduction leads to two intertwined problems. The first asks for explicit module categories over $\mathcal{T}$. The second asks for an explicit description of the morphisms between them, namely the right exact $\mathcal{T}$ -module functors. My current work addresses both, with the second serving as the conceptual center of my dissertation.

For each divisor $d \mid \ell$, let

$$\mathcal{M}_d := \text{Rep}(Q_d, I^\ell),$$

where Q_d is the oriented d -cycle and I^ℓ is the ideal generated by all paths of length ℓ . These nilpotent cycle categories are the basic quiver models of the project. The first step is to show that the Taft coproduct itself determines an explicit action of $\mathcal{T}$ on each $\mathcal{M}_d$. This is not an abstract existence statement: the action is written directly in terms of bundled weight spaces and the coproduct formula, that is, the components of the object $T \triangleright M$ are written in terms of the components of M and the weight spaces of T and the quiver structure maps $a_i^{T \triangleright M}$ are given by explicit formulas in terms of x , g , and a_i^M .

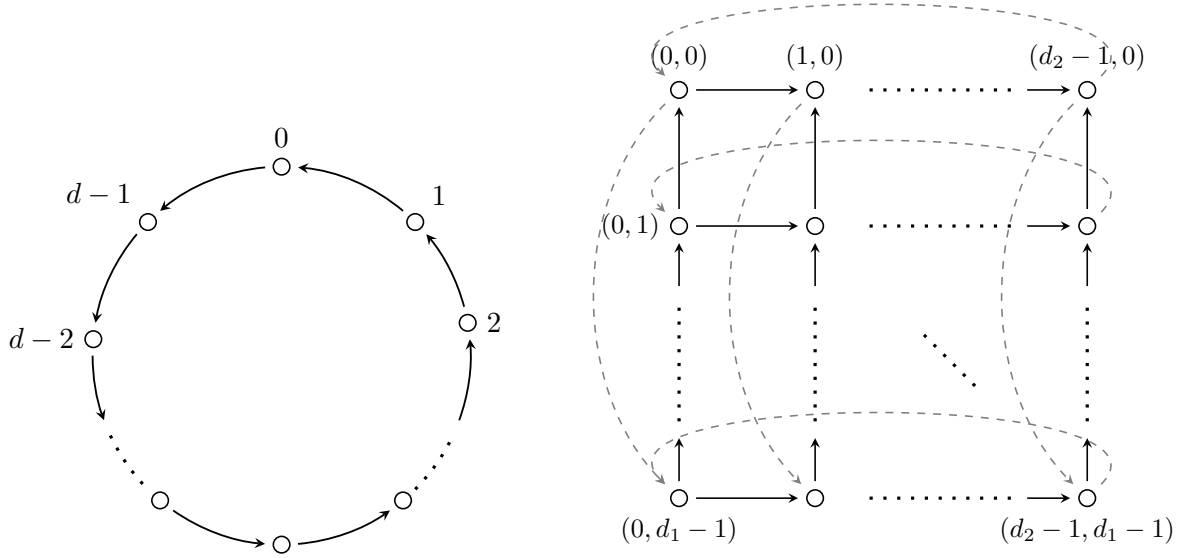


Figure 1: The d -cycle quiver Q_d and the $(d_2 \times d_1)$ -torus quiver.

Theorem 1 (Explicit standard module categories). *For each divisor $d \mid \ell$, the category*

$$\mathcal{M}_d = \text{Rep}(Q_d, I^\ell)$$

admits a canonical structure of exact indecomposable left $\mathcal{T}$ -module category. Moreover, every exact indecomposable left $\mathcal{T}$ -module category is equivalent, as a $\mathcal{T}$ -module category, to $\mathcal{M}_d$ for a unique divisor $d \mid \ell$.

This produces an explicit quiver-theoretic realization of the standard exact indecomposable left $\mathcal{T}$ -module categories, and the resulting family agrees with the existing classifications in the literature [Mom10, EO04].

This first result shows that the module categories themselves can be written in a concrete way. The action is visible at the level of quiver data. In effect, a tensor-categorical action is translated into a formula that can be manipulated directly. The next problem is to understand right exact $\mathcal{T}$ -module functors

$$\text{Fun}_{\mathcal{T}}^{\text{rex}}(\mathcal{M}_{d_1}, \mathcal{M}_{d_2}).$$

General categorical Morita theory says that right exact functors are represented by kernels in Deligne products. Here that means passing to objects of

$$\mathcal{M}_{d_2} \boxtimes \mathcal{M}_{d_1}^{\text{op}}.$$

At first sight this is still too abstract for computation. A major part of my current work is to rewrite these Deligne products in quiver language, obtaining a mixed torus-grid model in which horizontal and vertical directions record the target and source cyclic data. This gives an explicit ambient category for all right exact functors between the standard module categories.

The crucial issue is that not every kernel in this ambient category comes from a $\mathcal{T}$ -module functor. The extra requirement is equivariance with respect to the $\mathcal{T}$ -actions on source and target. In the Taft setting, that equivariance can itself be made explicit.

Balanced kernels. A *balanced kernel* is a pair (K, β) consisting of an object

$$K \in \mathcal{M}_{d_2} \boxtimes \mathcal{M}_{d_1}^{\text{op}}$$

together with a natural family of isomorphisms

$$\beta_X : X \triangleright K \xrightarrow{\sim} K \triangleleft X \quad (X \in \mathcal{T}),$$

compatible with the tensor product and the unit object. The resulting category is denoted

$$\text{Bal}_{\mathcal{T}}(\mathcal{M}_{d_2} \boxtimes \mathcal{M}_{d_1}^{\text{op}}).$$

This definition isolates the precise extra structure needed for a kernel to encode a $\mathcal{T}$ -module functor. The point is not merely formal. In the mixed-grid picture, the balancing attached to the one-dimensional character and to the standard two-dimensional indecomposable forces an explicit interaction between the two quiver directions. The remarkable consequence is that the apparent two-dimensional kernel calculus collapses to a single cyclic direction.

Theorem 2 (Classification of module functors by balanced kernels). *For divisors $d_1, d_2 \mid \ell$, right exact $\mathcal{T}$ -module functors are classified by balanced kernels. More precisely,*

$$\text{Fun}_{\mathcal{T}}^{\text{rex}}(\mathcal{M}_{d_1}, \mathcal{M}_{d_2}) \simeq \text{Bal}_{\mathcal{T}}(\mathcal{M}_{d_2} \boxtimes \mathcal{M}_{d_1}^{\text{op}}).$$

Moreover, if $r = \gcd(d_1, d_2)$, then the balancing relations force a collapse from the mixed torus-grid to a single oriented cycle of length r , yielding an explicit classification in terms of the nilpotent cycle category $\mathcal{M}_r$.

Conceptually, this is the center of the project. It shows that tensor-categorical linearity is not an invisible coherence condition in this example: it can be read as a concrete quiver relation. Once written in that form, it rigidifies the ambient kernel calculus enough to make classification possible.

Just as importantly, the kernel viewpoint is not merely classificatory. Module functors compose, so their kernels carry a corresponding convolution law. In the endomorphism case this produces a monoidal structure on the category of balanced endokernels. This places the classification inside a broader structural framework: the project is not only about listing module categories and module functors, but also about making their internal calculus explicit.

Taken together, these results support a broader research philosophy. The quiver model is not a decorative translation of the Taft category into graph language; it is the mechanism that turns categorical Morita theory into something concrete. That makes it possible to ask sharper questions about structure, comparison, and generalization. The future directions below are designed to push on exactly that point.

3. Future Directions

The next stage of my research is to develop the present Taft results into a broader quiver-theoretic program for non-semisimple tensor categories. The projects below are organized to move in four complementary directions: isolating the minimal combinatorial core, placing the current results in a conceptual 2-categorical framework, extending the method to new tensor categories, and extracting algebraic invariants from the kernel calculus.

Project 1. Skeletal quiver models for tensor-categorical actions

The current construction is carried out in representation categories of bound quivers. A natural next step is to ask whether the essential argument already exists one level lower, at the level of skeletal path categories and relations.

Goal. Rebuild the Taft module-category action and balanced-kernel formalism directly from path categories, additive hulls, and relations, so as to identify the minimal combinatorial input needed for the classification.

This project asks how much of the current theory is genuinely categorical, and how much is already forced by the underlying quiver combinatorics. If successful, it would isolate the core mechanism behind the explicit action and the mixed-grid collapse, while clarifying exactly where abelian or finiteness hypotheses first become essential. It would also produce a cleaner blueprint for transport to other examples: instead of repeating a large amount of ambient categorical machinery each time, one could begin with skeletal data and then pass to representation categories only when necessary.

This direction connects tightly to my current work because the formulas already suggest that the real structure lives at the level of paths and relations. It also complements the next project: once the combinatorial core has been isolated, one can ask how that core fits into a larger bicategorical framework.

Project 2. A bicategorical Morita theory of balanced kernels

The classification of module functors by balanced kernels naturally suggests a 2-categorical interpretation. Classical Morita theory organizes algebras, bimodules, and bimodule maps into a bicategory. Finite tensor categories, module categories, module functors, and module natural transformations admit an analogous higher-categorical organization.

Goal. Formulate the bicategory whose objects are the standard Taft module categories $\mathcal{M}_d$, whose 1-morphisms are right exact $\mathcal{T}$ -module functors, and whose hom-categories are modeled explicitly by balanced kernels.

The conceptual point is that my current results do more than classify individual functors: they identify a concrete model for the morphism categories between module categories, together with a composition law coming from kernel convolution. Making this bicategorical structure explicit would

show precisely how the balanced-kernel category sits inside categorical Morita theory. It would also clarify which parts of the present story depend specifically on the Taft algebra and which belong to a more general pattern of equivariant kernel calculi.

This project is especially appealing because it turns the current work from a family of classification statements into a structured piece of higher representation theory. It also creates a natural language in which Project 1 and Project 3 can interact: the former isolates the combinatorial inputs to the bicategory, and the latter asks in which other tensor categories similar bicategories can be described explicitly.

Project 3. Extending the quiver-theoretic method beyond the Taft category

The Taft tensor category is a particularly accessible non-semisimple example, but it is not the only one whose Hopf-theoretic structure hints at quiver models. The broader challenge is to identify which features of the present argument are special to Taft and which are robust.

Goal. Determine structural criteria under which a finite tensor category admits explicit quiver models for its standard module categories and for the kernels of its equivariant functors.

Promising test cases include pointed Hopf algebras and related tensor categories in which grouplike and skew-primitive data still organize representations into periodic or graded combinatorial patterns. The central question is not simply whether one can repeat the Taft argument verbatim, but whether there is a general mechanism by which tensor-categorical compatibility can again be rewritten as a concrete quiver relation.

This project matters because it pushes the current work from an explicit example toward a reusable method. The Taft category is the right place to learn how the mechanism works; broader families will test whether the mechanism is truly structural. In that way, the present results serve both as a solved case and as a template.

Project 4. Composition, convolution, and decategorification

A final direction is to study the algebraic shadow of the balanced-kernel calculus. Since module functors compose and balanced endokernels carry a monoidal structure, their Grothendieck groups and decategorified invariants should encode meaningful information about the Morita theory of $\mathcal{T}$.

Goal. Analyze the monoidal category of balanced endokernels through convolution and decategorification, with the aim of extracting computable algebraic invariants attached to Taft module categories and their equivariant endofunctors.

This project offers a bridge between the categorical and algebraic sides of the research program. On the categorical side, it refines the composition law already visible in the kernel picture. On the algebraic side, it asks what ring-like or representation-theoretic structure remains after decategorification. Such invariants could provide a compact summary of the interaction between module

categories and equivariant functors, and they may also give a language for comparison with more classical convolution constructions.

Within the larger program, this direction plays an important balancing role. Projects 1–3 focus on constructing and conceptualizing explicit models. Project 4 asks what concrete algebraic data those models produce. It therefore turns the current theory outward, making it easier to compare with neighboring areas of representation theory and tensor-categorical algebra.

These projects are closely related. Project 1 isolates the minimal combinatorial engine behind the current classification. Project 2 explains where that engine lives in the higher-categorical landscape. Project 3 tests how widely the method can be exported. Project 4 studies the algebraic structures produced by the method once composition is taken seriously. Together they define a coherent research program: to make categorical Morita theory over non-semisimple tensor categories explicit enough to compute with, conceptual enough to organize structurally, and flexible enough to extend beyond a single example.

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